WORKSHEET 2.4

Corners, Cusps and Breaks: When the Derivative Fails

Unit 2: The Derivative

Key ideas

- A function is not differentiable at a **corner**, a **cusp**, a **vertical tangent** or a **discontinuity**.
- Differentiable implies continuous. Continuous does *not* imply differentiable.
- A piecewise function is differentiable at a joining point when the pieces **meet** (continuity) and have the **same slope** there.
- To find unknown constants, write one equation for continuity and one for equal slopes.

Worked examples

Example 1 · A corner

Show that $f(x) = |x + 2|$ is not differentiable at $x = -2$.

Solution. For $h > 0$, $\dfrac{|h|}{h} = 1$; for $h < 0$, $\dfrac{|h|}{h} = -1$. The one-sided limits of the difference quotient are 1 and -1.

The limit does not exist, so $f'(-2)$ does not exist.

Example 2 · Checking a join

Is $f(x) = \begin{cases} x^2, & x \le 3 \\ 6x - 9, & x > 3 \end{cases}$ differentiable at $x = 3$?

Solution. Values: 9 and $18 - 9 = 9$, so f is continuous. Slopes: $2(3) = 6$ and 6.

Yes: the pieces meet with the same slope.

Example 3 · Finding constants

Find a and b so that $f(x) = \begin{cases} ax + b, & x < 2 \\ x^2 + 1, & x \ge 2 \end{cases}$ is differentiable at $x = 2$.

Solution. Slopes: $a = 2(2) = 4$. Continuity: $2a + b = 5$.

$a = 4$ **and** $b = -3$**.**

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Practice

CORE

1. At what value of x is $f(x) = |x - 5|$ not differentiable?

2. Is $f(x) = x^{2/3}$ differentiable at $x = 0$? Give a reason.

3. Is $f(x) = \begin{cases} 4x - 1, & x \le 1 \\ 2x^2 + 1, & x > 1 \end{cases}$ differentiable at $x = 1$?

4. True or false, with a reason: every continuous function is differentiable.

EXAM LEVEL

5. Is $f(x) = \begin{cases} x^2 + 2x, & x \le 0 \\ \sin(2x), & x > 0 \end{cases}$ differentiable at $x = 0$? Justify your answer.

6. Find a and b so that $f(x) = \begin{cases} ax^2, & x \le 2 \\ 4x + b, & x > 2 \end{cases}$ is differentiable at $x = 2$.

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7. At what values of x is $f(x) = |x^2 - 1|$ not differentiable?

8. Is $f(x) = \begin{cases} \sqrt{x}, & 0 \le x \le 4 \\ \frac{x}{4} + 1, & x > 4 \end{cases}$ differentiable at $x = 4$? Justify your answer.

Question 9 · Free-response practice

Let $g(x) = \begin{cases} x^2 - 4x, & x \le 1 \\ ax + b, & x > 1 \end{cases}$ where a and b are constants.

(a) Write an equation in a and b that makes g continuous at $x = 1$.

(b) Find the values of a and b for which g is differentiable at $x = 1$.

(c) With those values, write an equation of the line tangent to the graph of g at $x = 1$. What do you notice?

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Worked solutions

1. $x = 5$.
2. No. $f'(x) = \frac{2}{3}x^{-1/3}$ is unbounded near 0: the graph has a cusp.
3. Yes. Values: 3 and 3. Slopes: 4 and $4(1) = 4$.
4. False. $f(x) = |x|$ is continuous everywhere but not differentiable at 0.
5. Yes. Values: 0 and $\sin 0 = 0$. Slopes: $2(0) + 2 = 2$ and $2\cos 0 = 2$.
6. Slopes: $4a = 4$, so $a = 1$. Continuity: $4 = 8 + b$, so $b = -4$.
7. $x = -1$ and $x = 1$, where $x^2 - 1$ changes sign and the graph has corners.
8. Yes. Values: 2 and 2. Slopes: $\frac{1}{2\sqrt{4}} = \frac{1}{4}$ and $\frac{1}{4}$.
9. (a) $1 - 4 = a + b$, so $a + b = -3$.
   (b) Slopes: $2(1) - 4 = -2 = a$. Then $b = -1$.
   (c) The point is $(1, -3)$ and the slope is -2: $y + 3 = -2(x - 1)$, that is $y = -2x - 1$. This is the linear piece itself: the right-hand piece is the tangent line to the parabola.

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